

Tinnaphop Sonsang

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A Computer Engineering graduate with hands-on experience building end-to-end machine learning pipelines, from data engineering and model training to real-world deployment. Skilled in computer vision, data engineering, and full-stack web development, with a focus on production-ready systems.

EDUCATION

Walailak University

B.Eng. Computer Engineering and Artificial Intelligence

GPAX: 3.65 (First-class honors) | Graduation: 7 May 2026

Nakhon Si Thammarat

June 2022 – 2026

CERTIFICATIONS

EF SET English Certificate ([View certificate: cert.efset.org/en/gCMcLJ](https://cert.efset.org/en/gCMcLJ))

Awarded 3 Aug 2026

- Overall score: 56/100 — B2 Upper Intermediate (CEFR); Reading: C1 Advanced (64/100)

KEY SKILLS

- End-to-End AI Pipeline Design:** Built multi-stage computer vision pipelines combining object detection, depth estimation, and pose estimation for real-world deployment, including a YOLO-based detection model achieving mAP@50 of 0.955.
- Data Engineering & Unsupervised Learning:** Built scalable pipelines to extract and process 11+ audio features from Spotify tracks, applying KMeans, Gaussian Mixture Models, PCA, and UMAP for model selection, dataset deduplication, and image filtering.
- Full-Stack Web Development:** Astro, Next.js, React, TypeScript, Tailwind CSS on the frontend; PostgreSQL (Supabase), SQLite, Drizzle ORM on the backend; integrated real-time AI inference endpoints into production web applications.
- Tools & Productivity:** Figma, Adobe Photoshop, Canva, Microsoft Office Suite.

WORK EXPERIENCE

Intern Developer (Co-op/Internship) | ThaiDotRun, Bangkok

Sept 2025 – Apr 2026

Automated Runner Highlight Extraction System

- Designed an automated video clipping pipeline for running events using an object detection model (RT-DETR) and BoTSORT for subject tracking
- Implemented a custom spatial weighting algorithm to prioritize subjects closest to the camera center, and integrated OCR to extract runner bib numbers, dynamically replacing track IDs to determine precise start and end frames for automated highlight generation
- Achieved an average success rate of approximately 60% for automated highlight video generation, establishing a solid baseline for initial deployment and future optimization

Robustness Shoe Detection & Classification Pipeline

- Designed and developed an end-to-end deep learning pipeline for detecting and classifying brands from complex images and video footage
- Applied unsupervised learning for dataset deduplication to eliminate redundant training frames; used Grounding DINO to detect and localize runners and extract shoe crops for an ensemble learning pipeline trained on the deduplicated dataset
- Implemented depth-mapping and keypoint detection to assess image quality and enhance overall prediction reliability
- Successfully integrated an ensemble classification framework, providing a comprehensive foundation for automated quality control and data re-collection

PROJECTS

Complex-Valued Neural Networks: When Does Phase Actually Help?

July 2026 – Present

- Designed and implemented a complex-valued CNN from scratch (Cartesian weights, tied convolutions, shared-accumulator branching with per-neuron ModReLU) for speech classification on complex STFT spectrograms
- Diagnosed training instability via full covariance-based complex batch normalization (2×2 whitening) and Trabelsi-style weight initialization, isolating these from architecture choice
- Designed a controlled synthetic experiment with a custom weight-tracking diagnostic to test whether phase-dependent task structure, not model capacity, determines when complex representations help
- Found that even with parameter-matched capacity (~1.02×) and literature-standard initialization, the complex model underperformed a real-valued baseline by ~35pp, consistent with current literature on tasks lacking genuine phase structure

Madonna Discography Analysis

March 2026 – June 2026

- Developed an end-to-end music analytics pipeline applying unsupervised learning to uncover audio-based patterns across Madonna's discography, using a deduplicated set of Spotify audio features
- Addressed heavy right-skew in features such as instrumentalness, speechiness, and liveness via Yeo-Johnson power transformation prior to clustering, correcting an initial failure mode where outlier tracks were isolated into non-meaningful micro-clusters
- Benchmarked KMeans against a Gaussian Mixture Model at K=4, selecting the better-fitting algorithm via silhouette score and Davies-Bouldin index, and validated cluster stability using Adjusted Rand Index and bootstrap resampling
- Used UMAP for dimensionality reduction to visualize high-dimensional audio relationships, and auto-generated 4 "Sonic Persona" labels from per-cluster z-scores (e.g., Vocal-driven & Melodic) instead of manual labeling

Sitedotmoss (sitedotmoss.vercel.app)

December 2025 – Present

- Developed a product-level, full-stack interactive portfolio featuring advanced state management and real-time AI capabilities
- Architected the solution utilizing **Astro**, **React**, **TypeScript**, and **Tailwind CSS** for the frontend, **SQLite/Drizzle ORM** for the backend, and **Hugging Face Spaces** for AI inference
- Delivered a highly engaging, retro-futuristic user experience featuring cohesive Y2K/Frutiger Aero aesthetics, glassmorphism, and an intuitive tile-based structural design

Capstone Project (Thesis) — Detection of White Blood Cells in Raptors

August 2024 – August 2025

- Developed a YOLO-based object detection model trained on a custom dataset of 1,200 images to accurately identify six classes of white blood cells in raptors
- Architected a full-stack web application utilizing Next.js, React, and PostgreSQL (via Supabase) to deploy the model, enabling users to upload image batches and receive diagnostic predictions
- Achieved a mean Average Precision (mAP@50) of 0.955, and an mAP@50-95 of 0.742 across multiple IoU thresholds
- Manuscript based on this research is currently in preparation for peer-reviewed publication